

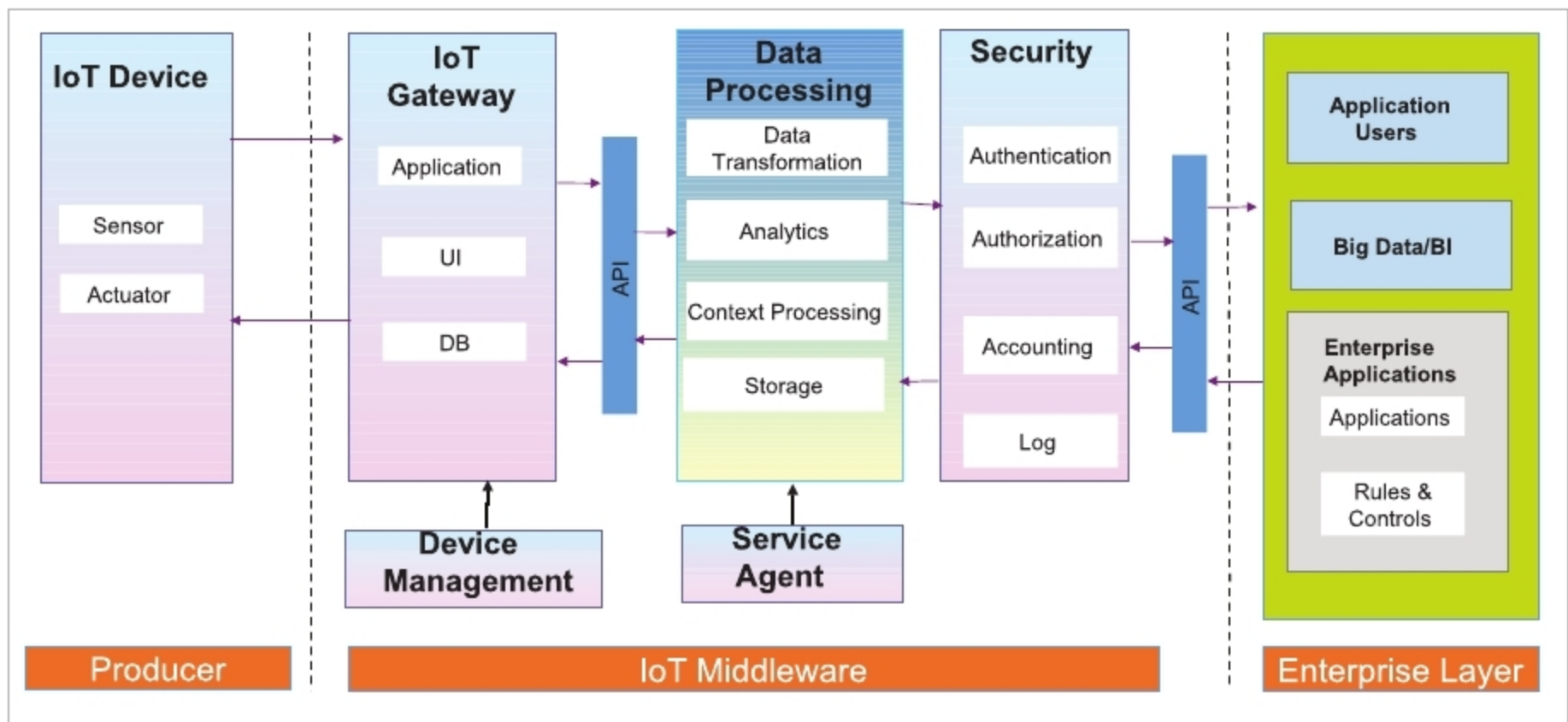


Figure 1: IoT middleware architecture

The IoT data stream normalises the data to a common format and sends it to enterprise systems. The database needs to perform the data transformation operations efficiently.

Middleware supports the authentication of users, organisations, applications and devices. It supports functionalities like certificates, password credentials, API keys, tokens, etc. It should also support single sign-on, time based credentials, application authentication (via signatures) and device authentication (via certificates).

Logging is necessary for both system debugging as well as auditing. Middleware manages the logging of system debugging and auditing details. It helps to track the status of the various services, APIs, etc, and administers them.

Key open source IoT middleware platforms

Holistically, an IoT implementation covers data collection and insertion through sensors as well as giving control back to devices. The different types of IoT middleware are categorised as:

- Application-centric (application and data management)

- Platform-centric (application enablement, device management and connectivity management)
- Industry-specific (manufacturing, healthcare, energy and utilities, transportation and logistics, agriculture, etc)

Selecting the right middleware during various stages of IoT implementation depends on multiple factors like the size of the enterprise, the nature of the business, the development and operational perspectives, etc.

The following are some of the top open source middleware platforms for IoT based applications.

Kaa is platform-centric middleware. It manages an unlimited number of connected devices with cross-device interoperability. It performs real-time device monitoring, remote device provisioning and configuration, collection and analysis of sensor data. It has microservices based portability, horizontal scalability and a highly available IoT platform. It supports on premise, public cloud and hybrid models of deployment. Kaa is built on open components like Kafka, Cassandra, MongoDB, Redis, NATS, Spring, React, etc.

SiteWhere is platform-centric middleware. It provides ingestion,

storage, processing and integration of device data. It supports multi-tenancy, MQTT, AMQP, Stomp, CoAP and WebSocket. It seamlessly integrates with Android, iOS, and multiple SDKs. It is built on open source technology stacks like MongoDB, Eclipse Californium, InfluxDB, HBase and many others.

IoTSyS is platform-centric and industry-specific middleware. It uses IPv6 for non-IP IoT devices and systems. It is used in smart city and smart grid projects to make the automation infrastructure smart. IoTSyS provides interoperable Web technologies for sensors and actuator networks.

DeviceHive is cloud agnostic, microservices based, platform-centric middleware used for device connectivity and management. It has the ability to connect to any device via MQTT, REST or WebSockets. It supports Big Data solutions like Elasticsearch, Apache Spark, Cassandra and Kafka for real-time and batch processing.

EclipseIoT (Kura) provides device connectivity, data transformation and business logic with intelligent routing, edge processing/analytics and real-time decisions.